Smart Plants — Quick Build Reference

Waveshare ESP32-C6 Zero · smart_plants_breakout_ws-board_rev2

Full instructions

1. Solder the PCB

Match orientation-sensitive components to the silkscreen. Resistors are not orientation-sensitive. Suggested order: resistors + diodes → JST connectors → pin sockets → capacitors + MOSFET.

Ref	Component	Value	Notes
R1, R2	Resistor	220 kΩ	Battery voltage divider
R3	Resistor	10 kΩ	MOSFET gate pull-down
R4	Resistor	22 Ω	MOSFET gate protection
R5, R6	Resistor	4.7 kΩ	optional, I2C pull-up
D1	Diode	1N4001	orientation sensitive, Flyback — stripe per silkscreen
D2	Diode	1N5817	orientation sensitive, Schottky — stripe per silkscreen
J3, J5, J6, J11	JST XH 4-pin	—	I2C connectors — need only the first two from the top
J9	JST XH 3-pin	—	Moisture sensor
J7, SW1	JST XH 2-pin	—	Battery, switch
J10	JST XH 2-pin	—	Pump
J4	JST PH 2-pin	—	USB-C power (2.0 mm pitch)
J1, J2	9-pin socket	—	Insert Waveshare board after soldering
C2	Capacitor	220 µF	orientation sensitive, Electrolytic — long leg to +
C3	Capacitor	22 µF	orientation sensitive, Electrolytic — long leg to +
Q1	MOSFET	IRLZ14	Flat side matches silkscreen

⚠️ Legacy PCB bug: on old PCB versions the moisture connector (J9) is labelled "D1-powered" — correct GPIO is 21. Ignore the label.

2. Prepare Sensors & Actuators

Moisture sensor cable — cut 3-core cable to length, crimp 3-pin JST XH on both ends.
Wire order: SIGNAL (yellow) — GND (black) — VCC (red). Discard the short cable from the sensor.
Insert sensor into housing, screw on lid.

Pump — solder 2-core cable to pump. Insert into housing, fit PG7 cable gland, crimp 2-pin JST XH on free end.

BME280 — solder pre-crimped cable: VCC (black) — GND (red) — SCL (yellow) — SDA (white).
Insert into Stevenson screen, thread cable through tube.

TSL2591 — same wire order as BME280. Wires away from sensor face.
Hot-glue sensor into housing (facing up). Glue half ping-pong ball into groove as diffuser.

> ⚠️ Pre-crimped cables: black = VCC, red = GND — reversed from convention. Double-check before soldering.

Switch — solder 2-core cable, crimp 2-pin JST XH. Wire order doesn't matter.

Battery pack — crimp 2-pin JST XH: VCC (black) — GND (red).

3. Assemble the Enclosure

1. Slot laser-cut panels together; add corner pieces *while* joining sides.
2. Drill 8 mm hole for sensor cables if not pre-cut.
3. Mount BME280 through lid hole (distance piece + locking piece). Hot-glue TSL2591 housing to lid. Keep sensors as far apart as possible.
4. Insert switch and USB-C connector into housing. Feed pump and moisture cables in.
5. Insert battery pack into PCB mount. Screw PCB to mount. Insert Waveshare board into socket.
6. Velcro PCB mount to housing interior. Connect all components to labelled PCB connectors.
   BME280 → **I2C D2-powered** | TSL2591 → **I2C D3-powered**

4. Flash the Firmware (from the browser)

Use **Chrome, Edge, or Opera** and a USB-C **data** cable.

1. **Flashing mode:** unplug → hold **BOOT** → plug in → release after ~2 s (the firmware sleeps quickly and drops the USB port).
2. Open **vektorious.github.io/hpi_smart_plants**.
3. **Connect** → pick the `USB JTAG/serial debug unit` port → **Install Smart Plants Firmware**.

> Building from source instead? See `firmware_from_source.md`.

5. Configure (setup portal)

On first boot the device opens a Wi-Fi hotspot **`SmartPlant-XXXXXXXX-Setup`** (unique per board, no password).

1. Connect to it; the setup page opens automatically (or browse to `192.168.4.1`).
2. Set **Wi-Fi**, and optionally device name, sensors/pump, and calibration. The API key is pre-filled. **Save.**

> Reopen the setup page later to change anything: **double-press the reset button**.

6. Check & Calibrate

Open **Live Sensor Readings** in the portal:

- Values refresh every 10 s; the Wi-Fi badge shows the connection.
- **Send to API** → green **200 OK** confirms Wi-Fi + key + server.
- A sensor showing **"not detected"** → check its cable / wire order.

Calibrate moisture from the same page — read the *moisture voltage*: in open air = *dry voltage* (0 %), prongs in water = *wet voltage* (100 %). Enter both in the portal and save — no reflash. Defaults: wet 0.60 V, dry 2.45 V.

7. Finish & Deploy

1. Click **Finish setup** → device takes a reading and starts its hourly cycle.
2. Confirm it appears on **plants.makeruniverse.de**.
3. Disconnect USB, flip the power switch on, screw the lid on.

> 💡 A single reset triggers an immediate reading; a double-press reopens the setup page.